DO NOW · AO1 REVISION

Red-shift

Space physics
Paper 2 · Separate Physics only
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**What is red-shift?**	The increase in the observed wavelength of light from galaxies that are moving away.	☐
2 ★	**Explain why light from distant galaxies is red-shifted.**	The galaxies are moving away from us.	☐
3	**Describe how the red-shift of a galaxy depends on its speed.**	The faster it is moving away, the greater the red-shift.	☐
4	**Describe how the speed of a galaxy depends on its distance from us.**	The further away it is, the faster it is moving away.	☐
5	**Describe the Big Bang theory.**	The Universe began from a very small, hot, dense region and has been expanding ever since.	☐
6 ★	**Explain why red-shift is evidence for the Big Bang theory.**	The galaxies are all moving apart, so they must once have been close together.	☐
7	**Name two things about the Universe that scientists still do not understand.**	Dark energy and dark matter	☐
8	**State what observations of supernovae since 1998 suggest about the Universe.**	Its expansion is accelerating.	☐
9	**State what happens to the observed wavelength of light from a galaxy moving away from us.**	It increases.	☐
10	**State what the red-shift of distant galaxies provides evidence for.**	That the Universe is expanding (the Big Bang theory).	☐

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